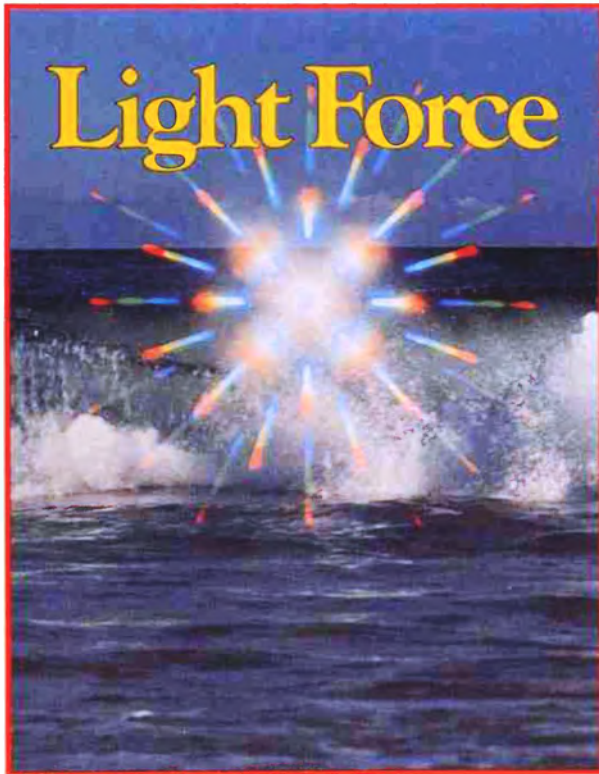


REJUVENATING THE BODY

**THROUGH FASTING WITH
SPIRULINA PLANKTON**



DR. CHRISTOPHER HILLS, Ph.d., D.Sc.

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Through Fasting With Spirulina Plankton

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REJUVENATION OF LIFE

For years mankind has sought for an elixir of life, a magical medicine that would give us eternal life. Sages researched all sorts of potions which put their consciousness in different states. This they called "soma", and sang its praises in the ancient scriptures. The ancient Chinese believed, like the alchemists, that the attainment of physical immortality initially was possible by taking some elixir which showed them the Divine Light. Chinese alchemy was concerned with rejuvenation and potency whereas western alchemists were mainly concerned with finding the philosopher's stone in the mysterious substance which can turn lead into gold.

The elixir of eternal youth, however, was not achieved by means of magical potions but by the purification of our body and its consciousness so that the subtle body, sometimes called the "double" or astral body, could travel or enter into other domains of experience. The lighting of the inner fires and the harnessing of the generative force was achieved through proper breathing and fasting in order to awaken this regenerative force which is stored at the base of the spine. Lighting the inner fire at the base of the spine sends the force up the spine to the top of the head to the unused portions of the brain and from there it descends back into all the cells and nerves of our body. The sages believed this was only achieved by certain mental practices as well as the physical practice of fasting.

The practice of purifying the body by purifying our desire for food and sexual appetite was amplified by what was called the immortal breath. In the process of immortal breathing we breathe into the body through the pores the subtle currents of life force as a stream of cosmic force entering the body at the heels or abdomen and travelling upwards to the brain. With every outward breath the

practice was to visualize a reverse flow of energy which took away with it all impurities. This led eventually to the production of a spiritual food or ambrosia in the center of the head which nourishes the seed or the nucleus of being itself. This divine embryo was considered the primal goal of all religious fasting because it reawakened the spark of the Godhead or the healing power which is at the center of every human being and all other beings.

The idea of ingesting certain foods and liquids which did the same thing on the physical level as the feeding of this embryo did on the spiritual level was not altogether an idea based on superstition. Just as we breathed in the divine fire, the “cosmic food”, to clear and purify our mind, so did they reason that we could take in pure nourishment that would wash away the poisons accumulated in the body through our ignorance of how to live in tune with the eternal laws of life.

Therefore fasting has a long history in the achievement of a superhuman energy or force of renewal. It is coupled with the taking of proper exercise such as yoga, jogging or martial arts in order to stimulate the organs of elimination and to provide a balance between the physical, mental and spiritual life.

Anyone who does fasting, jogging or yoga will tell you that the motive for doing it must be more than just losing weight and having a good body. There is also a feeling which goes with a healthy body or the practice of some discipline which we get as a reward for accepting and mastering the challenge we set ourselves.

The challenge of setting ourselves the goal of good health gives us the same satisfaction a gardener gets when he sets

out to grow some prize flowers, or the satisfaction of a musician who sets himself the challenge of learning to play an instrument well. There are just no shortcuts, but if you know how to do something and what ingredients, fertilizers, tips or kinds of soils to use, then you will get good flowers. Similarly if you learn music from a master of the art you will be able to perform reasonably well.

My other books, *Nuclear Evolution: Discovery of the Rainbow Body* and *Rise of the Phoenix*, are intended to fulfill the needs of anyone wishing to embark on the challenge of purifying their consciousness and awakening the inner fire through knowing the structure of the subtle instrument of the mind. This book on Spirulina plankton and fasting is intended to offer the physical challenge of purifying the body of all poisons through enhancing the process of elimination.

If your organs of elimination are sluggish or even diseased it may take time. You may expect a miracle because the human body is a miracle of sinew, blood and bone that is hard to match, but don't expect a miracle by tomorrow morning. Take time and do it properly and without a great deal of fanaticism. Follow these instructions carefully and then write me a letter telling me how you were helped or not helped. If you were not helped be sure to read these instructions again carefully before you write me a letter, because if you have missed anything or skipped any of the essential points, or even tried to substitute some pet idea of your own which changes these instructions in any way, then you can guess that I will not want to answer your letter.

I am myself a picture of perfect health and according to my students I get younger looking every day. Yet I do not

practice any particular faddist religion or method. I just automatically practice tuning in to the things you will find in this book. Like a wild animal eats grass to cure itself of colic, I eat certain ways, do heavy garden work or do running or take hot mineral baths whenever my system needs it. I take the minimum of exercise but I do it, because if I didn't do the minimum I would get sick. I do the minimum of fasting but when I need it I do it properly, not fanatically, and without cheating myself of the results.

The system I have outlined herein is the minimum for good health and if you do more or become a regular practitioner and do more than the minimum you will have more health and fitness than you know what to do with. What jogger does not want more energy, what spiritual athlete does not want more cosmic power? What sick person does not want the power to heal themselves?

I believe the Spirulina plankton is the "manna from heaven" which kept the Israelites alive for so many years among the rock pools of the desert. Including it in a fasting regimen is a way to get to a purified and healthy body in tune with cosmic laws. It is a first step in discovering the divine light and the power of self-rejuvenation.

THE WONDER VEGETABLE "MANNA FROM THE SUN" SPIRULINA PLANKTON

Fifty years of research has revealed the most nutritious food on the planet—Spirulina plankton.

Spirulina is a small alga growing in alkaline water. It is a complete vegetable protein cultivated in hygienic tanks and ponds under the latest scientific conditions.

Food From Sunlight gives the latest biological information on how to grow Chlorella and Spirulina as a way of solving the world food problem. Spirulina is even more nutritious and digestible than Chlorella and contains all the vitamins and minerals created by Nature needed for life. This plankton is the most concentrated form of any organic food and contains 65-71% protein, 10% carbohydrates, 7% fiber and 9% minerals and less fat than Chlorella (see following table.)

Spirulina has been known to mankind for centuries and many civilizations have found that all nutrients in Spirulina are abundantly photosynthesized due to the incredible reaction of this micro-vegetable to the sun's light. Whereas science has spent millions of dollars in scientific laboratories to harvest Chlorella efficiently for use in the space program and as a world food, the ancient Egyptians, Nubians and the people of Chad have eaten Spirulina safely for centuries because it is easily harvested. In 1963 I realized that it was dried algae which Moses found in the dried pools of the Egyptian deserts. I called Chlorella the new "Manna from Heaven" but when Spirulina was discovered I renamed it the "manna" because the desert is hot and alkaline. Spirulina grows at high temperatures of 30-35 degrees centigrade (86-95 degrees Fahrenheit). Because it floats on the surface in a thin layer and can be separated easily, it is more likely that Spirulina was the "manna" which gave the Israelites such good health.

Dr. Nakamura, an associate of mine, was appointed to direct the Microalgae Institute by the Emperor of Japan to solve the wartime food problem. Today Japan eats millions of pounds of algae and plankton because it is a total food source. Spirulina is the most light-weight concentrated nutrition so that only a small amount is needed. In animal

experiments it is found that one pound will feed a whole cow for a day when mixed in with the fiber or roughage. Only a small supplement of Spirulina is needed to get the equivalent of normal foods as the following comparison table for spray dried Spirulina shows*:

Food	% Protein	% Fat	% Carbohydrate	% Fiber	% Mineral
Spirulina	65-71.00	6.00	10.00	7.00	9.00
Chlorella	53.75	6.31	19.28	10.33	10.31
Cleaned Rice	7.48	0.94	90.60	0.35	0.72
Wheat	10.87	2.32	82.65	2.42	1.85
Barley	11.57	2.20	77.82	5.74	2.78
Soybean	28.99	19.93	30.93	5.11	5.58

WHAT SPIRULINA CAN DO FOR YOU

Spirulina is the greatest source of natural nutrition known to mankind providing a source of quick energy and protecting your cell life and endurance. As a source of nutrition during fasting and dieting it is excellent because it helps cleanse the intestinal tract as well as relax the smooth muscle of the bowels. The pigments in spray-dried Spirulina, namely Chlorophyll, Xanthophyll, Carotene, Phycocyanine, also interact with, and help produce the green pigments produced by the spleen and liver such as Bilirubin (bile pigment) used in the purification of the blood and detoxification of bacterial and mineral poisons. According to Merck's Manual the Bilirubin excreted in bile is converted to chromogens by intestinal bacteria.

Detoxification of the blood is achieved by balanced

* For figures of protein and vitamins in the fresh wet plankton, see *Food From Sunlight*, published by University of the Trees Press, 1978. Figures for heat-dried Spirulina are sometimes higher and sometimes lower for different components.

production of enzymes. Some liver cells over produce or leak enzymes and discharge them into the blood causing poor metabolism, poor excretion of bile into the intestine. Overretention of these pigments causes poor health and liver complaints. The pigments in Spirulina plankton and the amino acids are naturally available as a vitamin supplement and are excellent sources of enzymes for balancing metabolism and digestive health. Because the body of its cell is not enclosed in a mucous membrane as algae usually are, Spirulina is more highly digestible than Chlorella. When Chlorella is grown in certain acid/alkaline conditions the cell wall becomes hard and highly indigestible to certain people, hence the FDA has limited its use as a food. They are not against the Chlorella as the second most nutritious food in the world, but not everyone can tolerate the digestibility factor which is always dependent on how it is grown. Spirulina has none of these assimilation difficulties and contains 62-71 percent protein when dried as compared to 40-50 percent in dried Chlorella (production of Chlorella protein can go to 55 to 60 percent if we can accept a hard cell wall), compared with protein content of 39 percent in soybeans (miso) or 18-20 percent in beef, eggs or fish. By comparison rice is only 7 percent protein.

WHY HEALTH EXPERTS ARE SWITCHING TO SPIRULINA

Existing records reveal that the Aztec civilization reached a high intelligence in olden times, which may have been based on Spirulina. In the area which is now Mexico the Aztecs collected Spirulina and used it for making bread and preparing health-giving soups. Although this custom did not survive in the present-day Mexico the government there has now approved the Spirulina from Lake Texcoco as an acceptable food for nutrition. Several certified tests

have been done by commercial Japanese and American laboratories on Spirulina and those tests show already that Spirulina, dried by natives of Africa and Mexico, is a non-toxic food even when eaten in large quantities over a long period.

The Spirulina plankton cells are soft and group into filaments 100 times larger than Chlorella cells. Also, the taste is much better. Most Spirulina is spray-dried but the drum heat-dried variety turns brown and is almost identical in taste to Japanese Nori which is a delicacy wrapped around rice and available in gourmet delicatessens. The Spirulina grows in such high alkaline conditions that it becomes highly resistant to disease, other microorganisms and natural enemies.

HOW TO USE SPIRULINA PLANKTON

For people who are on a cleansing diet or fasting, the replacement of nutrients and the detoxification of the blood is important. Here is one short method out of the many which follow in this book, that people have tried with success.

Make a drink with:

the juice of one lemon

one tablespoon of maple syrup

Drink regularly as needed along with other juices or water, substituting four 500 mg. tablets of plankton for each meal.

The average person on this diet has lost up to 10 lbs. per two weeks.

ONLY ONE CALORIE PER CAPSULE (No sugar, no starch, no preservative)

People who have used Chlorella and had success with the marvels of this wonder food will absolutely recommend spray-dried Spirulina plankton as the vitamin food for the

new age. It is the ideal space food since man can live on small quantities of it for years and years.

Another way to use the plankton is to mix it uncooked with other foods like miso. Since Spirulina at present is more like a vitamin and still expensive to produce, the plankton can be extended by adding other sea kelp to it, but for cooking purposes you need to buy pure Spirulina plankton in powder form, in $\frac{1}{4}$ lbs. or 2 oz. containers for combination with other foods.

The Nori type of heat-dried Spirulina is not readily available and the green spray-dried from Japan is very expensive because so little of the concentrated plankton needs to be used. Tiny pills of it are made in Japan which also imports Spirulina from Siam and Mexico. Now that the production is being increased in Lake Texcoco, the Spirulina plankton is becoming available for the larger American market in the form of 500 mg. tablets. The powder, which promises to be the ideal survival food, is ideal for making up our own combinations and recipes.

One of the best combinations and extenders is miso (see *The Book of Miso*) which is made from fermented soybeans as follows:

- $\frac{1}{3}$ Spirulina plankton
- $\frac{2}{3}$ Hacho miso

This is very palatable and tastes like a cross between caviar and the bouillon cubes, and with the addition of sour cream on crackers tastes like a royal dish of Nori Supreme.

In Japan the fish fanciers know that this plankton hastens sexual maturity in some fish by as much as 60% and since this is a national tradition much of the Spirulina available goes to growing carp whose colors and size become wonderfully enhanced. Champion fish breeders pay large prices to get this nutritious food in Japan. Dogs and cats ravenously eat the Spirulina plankton mixed in with their normal food. This

conditions their fur and makes their coats shine. In Mexico City experiments have been done in breeding flamingos. The effect of feeding them Spirulina makes their plumage more brilliant and increases mating encounters. Many experiments have been done with mice and rats to measure the protein efficiency and to calculate their reproductive functions. The effect on the hair of humans is the same as for dogs and cats, say the Japanese who are already connoisseurs of algae and plankton in all its forms. Spirulina to the Japanese nutritionists is the wonder food tonic remedy, like snake gall bladders are to the Chinese, or wheatgrass is to the health enthusiasts of America.

A ¼ lb. or 2 oz. bottle of Spirulina powder can be extended for cooking by mixing with 1 lb. rice or any roughage product to make the most highly nutritious cooked meals available. But the Spirulina should be always added *after* cooking, because the natural enzymes have never been heated beyond 75 degrees and therefore still contain the essential growth factors which are causing the Japanese health-conscious public to confirm it as a youth-giving supplement. Spirulina provides not only rejuvenating stamina and energy but regeneration of tissues through enhanced nutritional elimination of toxins. You can send for a cookbook on how to prepare different recipes and soups from the publishers.*

FOOD FOR JOGGERS

Anyone needing a quick source of energy and a high rate of metabolism can facilitate Nature's own methods by using this improved strain of aquatic algae plankton to help

* A Spirulina recipe book, prepared by gourmet Ceylonese cook Sonia Beasley, will be available by Fall 1980.

other proteins and vitamins to get assimilated in the body. This plankton vitamin supplement provides the perfect solution to the need for a balanced source of youthful energy and vitamins. For over five years the Mexican Olympic team has been training with Spirulina supplements and the results showed in their winning a gold medal in the 1976 Montreal Olympics and again another gold medal in the 1979 Pan American Games. Details of how to use Dr. Hills Spirulina plankton for regular jogging are given on pages 36-39.

THE AVAILABILITY OF SPIRULINA

At present no Spirulina or Chlorella is grown in any sizable quantity in the USA, although there are some factories in Siam and Japan mainly stimulated by Japanese technology. As president of the Chlorella International Union, which changed its name to Microalgae International Union in 1968, the author, along with Dr. Hiroshi Nakamura, the Vice President, has kept closely in touch with the development and research. The greatest hope is for increasing the production in Lake Texcoco where it grows naturally so that the price can be kept lower. There are indications that prices will gradually be reduced with greater demand volume so that Spirulina vegetable plankton can eventually be used as a regular cooking product like miso, rice or vegetables. But with popular demand the Mexican prices may get higher before the increased production brings them down. Local consumption is expected to increase dramatically and a large company, Aquacultura Nutricional de Mexico S.A., has just been formed with semi-government interests. A move is afoot in Mexico to put five cents worth of Spirulina into Mexican bread to increase the protein of the loaf by fifteen percent. This will increase production. In the meantime the greatest user is a Japanese company represented by the author.

CHEMICAL ANALYSIS OF SPIRULINA

BY UNITED NATIONS LABORATORIES

PHYSICAL PROPERTIES

Appearance:	Fine powder
Color:	Dark green
Odor and taste:	Mild, resembling sea vegetables
Bulk Density:	0.5 g/l
Particle Size:	9-25 microns

CHEMICAL COMPOSITION

Moisture	7.0%
Ash	9.0%
Proteins	71.0%
Crude fiber	0.9%
Xanthophylls	1.80g/kg of product
Carotene	1.90g/kg of product
Chlorophyll a	7.60g/kg of product

TOTAL ORGANIC NITROGEN 13.35%

Nitrogen from Proteins	11.36%
Crude Protein (%N x 6.25)	71.0%

ESSENTIAL AMINOACIDS

Isoleucine	4.13%
Leucine	5.80%
Lysine	4.00%
Methionine	2.17%
Phenylalanine	3.95%
Threonine	4.17%
Tryptophan	1.13%
Valine	6.00%

NON-ESSENTIAL AMINOACIDS

Alanine	5.82%
Arginine	5.98%
Aspartic Acid	6.43%
Cystine	0.67%
Glutamic Acid	8.94%
Glycine	3.46%
Histidine	1.08%
Proline	2.97%
Serine	4.00%

VITAMINS

Biotin (H)	average	0.4	mg/kg
Cyanocobalamin (B ₁₂)	average	2	mg/kg
d-Ca-Pantothenate	average	11	mg/kg
Folic Acid	average	0.5	mg/kg
Inositol	average	350	mg/kg
Nicotinic Acid (PP)	average	118	mg/kg
Pyridoxine (B ₆)	average	3	mg/kg
Riboflavin (B ₂)	average	40	mg/kg
Thiamine (B ₁)	average	55	mg/kg
Tocopherol (E)	average	190	mg/kg

MOISTURE

7.0%

ASH

9.0%

Calcium (Ca)	1,315	mg/kg
Phosphorus (P)	8,942	mg/kg
Iron (Fe)	580	mg/kg
Sodium (Na)	412	mg/kg
Chloride (Cl)	4,400	mg/kg
Magnesium (Mg)	1,915	mg/kg
Manganese (Mn)	25	mg/kg
Zinc (Zn)	39	mg/kg
Potassium (K)	15,400	mg/kg
Others	57,000	mg/kg

NUTRITIONAL VALUE

Protein Efficiency Ratio (PER) of 2.2 to 2.6
(74-87% that of casein)
Net Protein Utilization (NPU) of 53 to 61%

(85-92% that of casein)

Digestibility of 83 to 84%

Available Lysine average 85%

NITROGEN FROM 1.99%

NUCLEIC ACIDS

Ribonucleic Acid (RNA) 3.50%

RNA = N × 2.18

Deoxyribonucleic Acid 1.00%

(DNA) DNA = N ×

2.63

TOTAL CARBOHY-

16.5%

DRATES

Ramnose average 9.0%

Glucane average 1.5%

Phosphoryled average 2.5%

cyclitols

Glucosamine and average 2.0%

muramic acid

Glycogen average 0.5%

Sialic acid and others average 0.5%

STEROLS

Cholesterol 325 mg/kg

Sitosterol 196 mg/kg

32 mg/kg

Dihidro 7 Cholesterol

Cholesten 7 ol 3

Stigmasterol

others

CAROTENOIDS

4,000 mg/kg

α Carotene traces

β Carotene average 1,700 mg/kg

Xanthophylls average 1,000 mg/kg

Cryptoxanthin average 556 mg/kg

Echineneone average 439 mg/kg

Zeaxanthin average 316 mg/kg

Lutein and Euglenanone average 289 mg/kg

TOTAL LIPIDS

7.0%

Fatty Acids

Lauric (C₁₂) 229 mg/kg

Myristic (C₁₄) 644 mg/kg

Palmitic (C₁₆) 21,141 mg/kg

Palmitoleic (C₁₆) 2,035 mg/kg

Palmitolinoleic (C₁₆) 2,565 mg/kg

Heptadecanoic (C₁₇) 142 mg/kg

Stearic (C₁₈) 353 mg/kg

Oleic (C₁₈) 3,009 mg/kg

Linoleic (C₁₈) 13,784 mg/kg

Linolenic (C₁₈) 427 mg/kg

Others 699 mg/kg

Insaponifiable 1.3%

Sterols 325 mg/kg

Triterpen alcohols 800 mg/kg

Carotenoids 4,000 mg/kg

Chlorophyll a 7,600 mg/kg

Others 150 mg/kg

3-4 Benzpyrene 3.6 mg/kg

Toxicology

Nontoxic

Heavy Metals

Typical

Arsenic (as As₂O₃) 1.10 ppm

Cadmium (Cd) less than 0.10 ppm

Lead (Pb) 0.40 ppm

Mercury (Hg) 0.24 ppm

Selenium (Se) 0.40 ppm

WHAT FASTING AND DIET IS ALL ABOUT

Severe protein malnutrition is not only found in Africa and poor countries at starvation level but right here in the land of plenty. Certain fasting, vegetarian and macrobiotic diets often result in protein-deficient people who become susceptible to sickness.

When I went to Hunza in 1961 only a few people had been there and had written books about the marvelous vegetarian fruit and nut diet responsible for their great health and longevity. When I got there it was the beginning of the onset of winter and the passes froze up, so I could not get out until the next thaw. I could not get to Gilgit which was the nearest airport. So I stayed there and ate the local food myself and found they loved to eat chicken once a week and also yak beef. It was true that during the summer they ate nuts and fruit because the crop of these was coming in, but I found their health was due not so much to their diet of natural foods but to the way they lived. Actually they ate very little by our standards, as food was always scarce and it was hard to grow in the rocky dust they call soil there. If they did not compost every leaf and all organic matter they could get their hands on, nothing would ever grow there in sufficient quantity because the whole area is deficient in topsoil. They dig holes in the ground and fill them with compost like we would fill a pot with soil. From these meager foods they get the minimum amount of proteins, consisting of about an average 0.28 grams per pound of body weight per day.

The whole population of Hunza gets plenty of exercise walking up and down their mountain terraces from 4 to 5 a.m. in the morning when they begin their work in the field until dark falls and they go to bed. Early to bed and early to rise and hard work make them into a terribly hardy and

spartan people. Add the fresh air of the high Himalayan range of the Korakorum peaks at 8,000 feet and you have a natural refrigerator which inhibits disease. They work hard and sweat to get two crops a year before the hard winter sets in which freezes the soil. Dried fruit and yak beef and milk must last them until the thaw. They come in for breakfast after three hours of work in the field to break their fast.

Protein food required by the human system depends on its quality and the size and type of work a person does. People eating high protein foods need less grams per day. The normal diet for a woman who eats animal proteins will be about 51 grams if her weight is approximately 128 pounds. But some people cannot use all the protein they eat because they cannot assimilate it efficiently. Next to Spirulina plankton as the highest source of vegetable protein in the world is Chlorella algae. But Chlorella has a hard fibrous cell wall and is not easily digestible, so all its 58% protein is never fully usable by the body. It is a similar story with other proteins like beef, fish, eggs and milk. In fact some people cannot tolerate milk at all. So it depends on each person how much of a given protein is recommended. The remarkable factor of Spirulina is that it is almost totally usable and is the most easily digested of all proteins known to mankind.

Plant proteins on the whole are regarded as a lower protein source than beef and therefore a higher amount of total protein is called for, around 65 grams instead of 51 grams for beef, i.e., for the same 128 lb. person ($0.51 \times 128 \text{ lbs.} = 65 \text{ grams}$).

The fact that Spirulina has three times the protein value of beef, nearly twice the value of soybeans and nine times

the value of rice and in addition is fully usable protein makes Spirulina the wonder food of this age, because so little of it is needed. Only about a third of the protein from a mixed diet of animal and vegetable proteins is usable by our body while all the protein of Spirulina is usable when we eat it or drink it.

To calculate how much protein we need to eat we can multiply the usable protein allowance of 0.28 grams per pound of body weight which comes out to 35.8 grams per day for a person weighing 128 lbs. or 43.1 grams for a person weighing 154 lbs. A rough estimate of the total intake of protein (usable plus the protein wasted) for a person living on vegetarian foods can be found by dividing the body weight in half and making the result into grams. Since I weigh on average about 180 lbs. my protein allowance should be about 90 grams of vegetable protein per day of normal plant life. But because the super food Spirulina plankton is four times as potent* and is all usable protein I will only need 22.5 grams to be completely nourished. A person of 128 lbs. would only need 16 grams ($64 \div 4$) and a person of 154 lbs., 19¼ grams ($77 \div 4$), depending on how active or how stressful our lives are. There are good indicators about our health in the condition of our nails and our skin, the fineness of our hair and the color of our tongue, and we can determine whether we need more or less.

The following table shows what we have to eat from a single source of protein to supply our daily protein needs.

Body weight	2,000 calories 128 lbs.	2,800 calories 154 lbs.
protein total	35.8 grams	43.5 grams
meat	7½ oz.	9 oz.
fish	8½ oz.	10 oz.

* Spirulina is four times as digestible as beef proteins. It can also be potentized homeopathically to be twice as beneficial as a source of trace elements.

milk	5 cups	6 cups
eggs	6 eggs	7 eggs
dry beans	12¾ oz.	15½ oz.
nuts	12 oz.	14 oz.

This makes an average of about 56 to 65 calories for each gram of protein consumed.

In order to get enough protein to be completely nourished without increasing weight the calorie/protein ratio should not exceed an average of about 60 calories to 1 gram of protein.

The remarkable factor about Spirulina plankton is that 100 grams contains only 393.9 calories (for a calorie/protein ratio of 3.9 calories per gram of protein). So you could eat as much as you like and not get fat; in fact you will get less and less fat day by day while being properly nourished.

This makes Spirulina plankton the ideal food for fasting and dieting or for adding to your favorite foods without fear of exceeding your calorie allowance, and is immediately converted into much needed protein for stoking up the energy of the body.

UNDERSTANDING FASTING

Unless you *want* to fast there is no use going through one, because if your desire to eat is greater than your desire to fast then you will continue to abuse your body by sneaking little tidbits into your mouth and stomach. The psychological condition while fasting is as important as the physical one because the emotions control the glandular functions and enzymes which regulate the cell metabolism.

But it is easier to take a juice fast while mixing Nature's wonder vegetable protein plankton with our favorite vegetable

and fruit juices. This way we can lose up to a pound a day. The part of the digestive system that normally handles solids is given a rest and the intestines are purified and replenished with good clean healthy flora. The kidneys, skin and liver need help to unclog the accumulated poisons which deposit themselves in all our cells and organs. If Nature has taken your appetite away it is your body trying to tell you that you may need to fast. You may need to stop sending more and more food into an unresponsive system. The best way to get back a good healthy appetite is not to eat. Fasting does not make you weak unless you continue it for a whole month and even then you will only feel weak if you try to do hard physical work. If you have strenuous mental work you will find it will become easier with a fast.

Although I believe long fasts can be abused by those who do not understand fasting and therefore I do not recommend it for everybody, a short fast of a week and a few days more will do nothing but good for your whole being. After the first hunger pains are relieved by drinking juice or water you can go on for days and even months without eating and without getting weaker. The feeling of hunger only returns after a long fast when all the toxins and waste products have been eliminated. Nature will set the time for completing your fast because you will get very hungry at the end and you feel ready to stop because your body knows Nature's job is done. Only at the beginning, for the first two or three days, will you feel that feeling of exhaustion. After this initial feeling you become much more alert as previously consumed food is eliminated from the intestinal tract.

If you are not going to fast from solid foods completely then there is no sense in starting in the first place. All rationalizations that you must nibble on something to

survive are a self-suggestion that you are not really interested in cleansing yourself through fasting and that you love food more than you love perfect health. Taking a little food other than your plankton mixed with vegetable or fruit juices will only worsen the situation because your bowel will not have enough food to move the feces down the intestinal tube. By the second or third day all desire for food will go anyway so to eat solids will only delay this natural repulsion for food. You will begin to feel more energy as your fast goes into its 4th and 5th days. To control your food and appetite is a self-discipline which you will learn by fasting. You should never go back to completely filling the stomach even after fasting because $\frac{1}{4}$ of the stomach should always be left free for churning the food in the digestive process.

There might be a headache as a reaction to the release of toxic wastes and your joints may ache. If this happens it shows how much you need to fast or change your diet. You must stick to the resolve to fast and permit no excuse which allows you to eat any solid whatever. As soon as you flush out your system with juices and the chlorophyll in *Spirulina* plankton, the eliminative organs will release their toxic debris. Then all your headaches and other symptoms you feel from fasting will disappear and you will begin to feel very clean and fresh instead of sluggish.

Once the hunger returns at the end of your fast you must not rush to gorge yourself but gradually return to solids. At first keep to vegetables and fruits, and then later on you can have proteins like beans and nuts mixed with *Spirulina* plankton. Avoid fats, fried foods, starches and sugar when you break your fast. When fasting you must get plenty of rest both mental and physical and keep yourself warm.

Also when you take warm baths you cause your pores to open and eliminate toxins as you sweat under the water. Go to bed early and try to break old habits. This is good for you to resolve not to slip back into the same patterns that make you sick or unhealthy in the first place. If fasting gives you nausea or bad breath remember that is part of the release of toxins. Watch your tongue daily. It will be heavily coated while the eliminating is taking place. Continue to flush out the toxins with vegetable juice or fruit juices mixed with *Spirulina* plankton. When your tongue is clear you will feel hungry again and you will be ready to go on to some solid fruits and vegetables and, in a few days, take up a new diet of nuts, fresh fruits and vegetables, with an occasional animal dish if you are not a vegetarian.

Many vegetarians are food faddists who all have their own theories about putrefaction. It is true that meat does putrefy in the intestine but this may not be because of the meat but because of the unhealthy system. A healthy meat-eater's stools do not smell foul whereas an unhealthy vegetarian's will. In perfect health you can theoretically eat anything and process it well but some people are just not suited to eating meat and some people are not suited to a strict vegetarian diet. There is no hard and fast rule in spite of all the fads, but there is one way to test it. Try being a vegetarian for a few months and if you have more vigor, feel clearer and your bowel is sweet and regular you will know the answer as to whether you are a vegetarian or not. Do not follow a fad but use common sense.

The amount of adrenalin you pump into your own blood when you get angry, even when you don't show it, is far more than the cow releases into its blood when it is killed. Its blood is drained after the slaughter anyway so the argument that you get more aggressive from meat-eating is

totally false.

I have known many meat-eaters who do physical work and they are not aggressive and I have met many self-righteous and opinionated, internally aggressive vegetarians. So you must think for yourself and try out your diet for yourself. If you are horrified at having to kill animals so you can eat meat then you can easily become a vegetarian on those grounds alone, and this is when you will need to get the extra protein from Spirulina plankton. People who have been big meat-eaters find it hard to get the same proteins and feeling of being “beefed up” from vegetables. By seeking out high-protein diets in plant materials like Spirulina they can compensate for this loss easily.

To understand fasting and diet properly the main thing is not to become a fanatic but to research your own personal diet because it makes common sense.

FASTING FOR HEALTH

Fasting with Nature's wonder food and mixing it with fruit and vegetable juices is the fastest and most natural way to flush out the system with liquids and chlorophyll without denying the body the needed nutrients for full and active metabolism.

The methods described in this booklet are simple ways of purifying the eliminative system and for having a healthy life.

For those with a weight problem they can reduce to their correct body weight quickly through natural means without any special dieting. Once they attain their goal they can stay on a fruit and plankton regime which will

supply all proteins and vitamins needed for health.

The use of special nutritious liquid foods or vitamins while fasting enables a clogged system to clear itself without feeling hungry or losing energy.

First we should clear the system properly through three days of pure fruit juice without taking any food except two tablets of Spirulina algae at each mealtime. The algae does not act as a solid food but actually works to purify the intestines through its chlorophyll content while supplying the full complement of amino acids necessary for healthy life.

Fasting is one of the best remedies available not only to the obese person but to anyone who has stomach troubles. It can also be used to increase our meditative and spiritual states of awareness. There are many examples in the Bible of fasting.*

Rejuvenation of body tissues and prolongation of our life was often the goal of ancient peoples through fasting. The practice was said to make the flesh of an older person as fresh and young as a child's. How does Spirulina plankton help in this process?

First we must understand that all healthy bodies create poisons in the normal process of living. These are the waste products of our cells and the unwanted parts of the foods we eat which are expelled by elimination. There are also many toxic materials in the modern foods we eat, the medicine we take, as well as the pollution in the waters and air which pile up in the sluggish system through inactive elimination. The secret of good health is in an efficient

* Luke 5:33, Sam. 12:16, Ex. 34:29, I Kings 19:8, Mat. 4:2

elimination of waste.

One of the worst inhibitors of our eliminative system is our mind and its negative thoughts. A clogged mind will invariably clog the movements of the smooth muscle of the intestine just as a lazy mind will make our whole body inactive and heavy. If we do not get this under control we will not only die early but our body will be always loaded with toxic material which will affect the workings of our brain. Therefore always drink plenty of liquids not only while fasting but also as a regular practice.

Toxic blood is passed through the special cells of the liver for clearing and since blood is the carrier of nutrients to all parts of the body and brain it is important to understand that the presence of pigments such as chlorophyll and the natural chemicals in the blood do affect the correct functioning of all other organs of elimination such as the bowels, kidneys, lungs, and skin.

When all these organs are in perfect balance the person remains in perfect health. Only when toxins cannot be handled by our eliminative system do the signs of ill health occur. There are always signs which indicate the start of disease or toxemia.

There is an inner intelligence* acting on its own which attempts to bring on warning signs of disease so as to get rid of poisons accumulating in the body. Colds, diarrhea, fever, etc., are Nature's ways of eliminating toxins which cannot be removed by normal elimination. Automatically we do not feel hungry when the body intelligence wants to

* A friend of mine, Dr. Oscar Morphis of the Oncology Associates in Texas, calls it *intelligence*.

eliminate its accumulated toxins. Animals do not eat while sick. Therefore fasting with Spirulina, while providing the essential amino acids, not only helps Nature's own process but is the most efficient method of correcting imbalance.

The ingestion of small amounts of Spirulina plankton (spray dried to remove the moisture from it) has an immediate effect on the muscle tone during periods of fasting. The following benefits show that fasting: (1) gives the vital organs a complete rest; (2) stops the intake of food which may decompose in the intestines and poison the body; (3) empties the digestive tract and disposes of putrefactive bacteria; (4) gives the organs of elimination an opportunity to catch up with their work and promotes elimination; (5) re-establishes normal physiological chemistry and normal secretion; (6) promotes the breaking down and absorption of exudates, deposits, diseased tissues and abnormal growths; (7) restores a youthful condition to the cells and tissues, and rejuvenates the body; (8) permits the conservation and re-canalization of energy; (9) increases the powers of digestion and assimilation; (10) improves function throughout the body; and (11) clears and strengthens the mind.*

HOW TO FAST

It is best to go on a fast gradually, cooperating with the internal intelligence of the cells so that they are not shocked by extremes. Therefore the following steps should be taken daily:

The first day there should be plenty of fresh juices but

* These are given by Dr. Hereward Carrington, Ph.D., noted researcher into parapsychological states.

no solids. With the juices one can ingest the Spirulina plankton to line the wall of the stomach and intestine with the nutritious form of chlorophyll and other pigments provided by blue-green algae. There is no limit to the amount one can eat in its pure form as it is completely digestible. Unlike Chlorella which has a hard cell wall and is often totally indigestible to some people, the Spirulina plankton is easily absorbed. Therefore the Spirulina can be mixed as powder with fruit juices or just taken in 500 mgs. tablet form. Two tablets (or 1000 mgs.) is a normal amount with a juice drink for fasting but any amount up to $\frac{1}{4}$ pound could be taken without harm at each mealtime, although this would not be advisable. Do it gradually by building up the system to adapt to the richness of this new wonder food by increasing the amounts daily.

During the second day of the fast you may add a little solid fruit in addition to the juices. This is to aid the elimination of any solids still stuck in the eliminative system. The fruit should be chewed very well, mixed with plenty of saliva thoroughly before swallowing. It is important for people using Spirulina plankton as a normal diet food to chew all other foods finely to allow full digestion to take place.

During the third day of the fast the fruit component can be eliminated until the final days of the fast when the normalizing of a regular diet of Spirulina mixed with other natural foods can take place.

On the fourth day we take fruit juices with at least four tablets or 2 grams of Spirulina plankton at meal times. Drink plenty of liquids in between meal times.

On the fifth day we again double the ingestion of the

amount of the cleansing diet to eight tablets of Spirulina plankton or 4 grams of the powder mixed with fruit juice.

The sixth day is the day when we begin to level off the peak of our intake of Spirulina plankton to that which we feel should be a natural amount for us. The temptation to eat something solid should be resisted during the fast because it stimulates the stomach into activity and causes it to churn whereas the Spirulina merely coats the walls and keeps it passive and relaxed. A relaxed stomach is part of the healing process and gives rest to the entire digestive system. It is much more beneficial to go on a complete Spirulina diet fast for the first week than to go on a partial fast of fruit or solids, but everything should be gradual.

On the seventh day we can begin to go back on to solid fruits and gradually bring down the intake of Spirulina from a minimum of 4 grams per mealtime to 2 grams reaching to 1 gram per meal on the eighth day. These amounts are minimal for the elimination effect of Spirulina.

If there should be headaches or stomach cramps during the fast this is not an indication that you must eat food but that the toxin accumulation is being thrown from the tissues and cells into the bloodstream and finds its way to the brain. This also affects the reflexes of the stomach, which has settled into a previous eating pattern which must be broken by the fast, rather than break the fast. Toxins cause pain and this is an indication that the body really needs to fast in order to get rid of toxins in the body. Healthy people do not get a headache merely by missing a meal; they only get a headache when the toxins reach the intestine or the bowel does not move. A bowel movement can instantly remove a headache.

If the bowel is constipated an enema made of the green spray-dried *Spirulina* mixed first as a paste and then introduced into a blender with distilled water should be used slightly warm to blood temperature. Do not use chlorinated tap water for your high enema as this will shock the sensitive walls of the intestine with chlorine, a very acid element which kills bacteria of all kinds. This is why it is put in drinking water, so you don't want it to kill the natural flora in your intestine which is building up through the use of *Spirulina*. An enema is good during a fast even if you do not have a headache but not essential, although it does speed the process of elimination. (See page 31 on for further information on the use of enemas with *Spirulina*.)

It is a mistake to set a rigid time limit to a fast, although a minimum of seven days should be allowed for purification of the digestive system and the cleansing of the blood. Your own feeling will determine how many days after the seventh day you should carry on the fast. Our body has natural signs which indicate the correct time to terminate a fast. The best time is when the tongue is clear and there are no white toxic adhesions and a natural hunger comes following the seventh day. If the tongue is not clear and a healthy pink color then prolong the fast until the hunger feeling coincides with a good clean color.

During the first three days while one is merely taking juices there will be a feeling of hunger while the system is cleaning itself. If one is sick, however, there may be no appetite during the first three days. Do not feed yourself to keep up your strength to fight the disease. This is a fallacy that food gives you strength to fight bacteria. If food cannot be processed and digested properly it acts as a poison rather than an energy source and merely clogs up the

system. Even vitamins can feed the bacteria as much as they feed the body, so should be used sparingly.

↓ After three days the hunger sensation in normal people disappears and it is quite easy to continue on a juice diet for several days or even weeks. I have myself fasted for 30 days without any feeling of hunger. But of course not everyone can take off time to meditate during a long fast and it is not wise to prolong a fast if you are moving around a lot or exercising or expending large amounts of physical energy.

If the toxins are all eliminated from the body the feeling will be one of tremendous clarity and lightness and we will feel our body has become free and clear. Then the hunger sensation will come back again and that is the time to break the fast. You should have no fear about losing nutrition or becoming too thin merely by not eating solid foods. Your Spirulina plankton is actually the most nutritious food in the world and supplies all the elements needed for health.

Once you break the fast you should not go immediately back to eating solid foods but gradually introduce them first as fruits and then blended vegetables which should be savored and drunk little by little to mix saliva with the juices and vegetables. While fasting, canned juices should be avoided. Preferably one should not go back to them after fasting but natural fruit and juices should always be used. If this is not possible because of cost or time then the canned juices should at least not be used during the period of the fast. All juices should be swilled around in the mouth to activate saliva. Do not drink them down straight. Remember, chewing them up in the blender or juicer is taking the place of your teeth and you must not omit the

most important step to good digestion, assimilation and excretion, which is the action of our saliva.

Several members of the University of the Trees staff who fasted extensively with Spirulina found that it took a few days to readjust once the fast was broken. Elimination needs to be encouraged at this time. They came off the fast by eating algae “mush”—a soup with a tomato juice–algae base thickened with a large quantity of bran—up to 3 tablespoons per serving, and seasoned to taste. Eating this once or twice a day for a few days stimulated elimination and reestablished healthy peristaltic action. Additionally, the bran served to sweep through the intestines, cleaning out any matter not fully eliminated during the fast.

You may find your intestines are sluggish to respond after the fast. It is helpful to take unprocessed bran mixed with soups or juices to provide the bulk to move through the intestines quickly. Drink a lot of water if you take bran. You can mix it with apple juice warmed or cold to make a delicious breakfast cereal and combine it with yeast, sesame seeds and raisins to taste. Bran is indigestible but it provides the bulk to move wastes quickly through the intestines and keep a healthy intestinal tract. You can mix bran into algae and miso soups, or algae and tomato juice for delicious soups.

Another way to stimulate the digestive process is to take acidophilus culture, between 1 and 3 tablespoons per day, to promote the growth of “friendly bacteria” to aid in digestion. After a long fast you can expect your intestinal flora to have changed. Taking acidophilus culture will help to normalize your intestinal flora.

If one does resort to an enema during a fast to hasten or assist elimination you must not overdo it and make a fad of it. Three nights in succession is enough for a healthy person. If one is sick, three more times every other night or every third night should be enough to flush the colon. Once normal body health has been restored the enema should be laid aside for Nature to carry out its own methods of elimination, and then infrequently a Spirulina enema can be used once a month.

Nine-tenths of human disease and the susceptibility to disease have their origin in the upper and lower intestines. Proper elimination of toxins through a clear alimentary system is the secret of health. To retain energy, cut down surplus body weight, clean out the system and at the same time use the most potent and nourishing form of food supplement known to mankind—Spirulina plankton—is the secret of health. Knowledge of how to clean the system is the essence of yoga. A mortuary in India examined the colons of 500 people who died and found the percentage of those which had colons greatly clogged with hardened fecal matter was 90% and only 10% had colons in good natural condition.

The color of the bowel movements is an indicator of our health. It does not always signify that we are healthy because we have a daily bowel movement because the upper part of the colon can still be impacted. Feces which are dark green, dark brown or black indicate they have been inside the colon a long time. The best color is light golden brown or yellow which shows they pass quickly through the bowel daily. If the fecal matter remains too long in its journey through the colon the toxic wastes contained in them are absorbed by the lower parts of the

system into the blood. This leads to poor skin, a furred tongue and bad breath in the morning. If we do not eliminate toxins our circulation of the polluted blood will be poor through the other organs resulting in rapid breathing, lack of memory, irregular heart action, dull eyes, impaired circulation to the brain leading to drowsiness and headaches.

Food which stays in the stomach, intestines and colon for longer than normal periods of 24 hours is apt to create fermentation and acid conditions. A sour stomach results from this. The liver and kidneys are forced to work overtime which results in susceptibility to infections.

In extreme cases Spirulina plankton fasting should be done with the use of enemas in order to purify the blood as soon as possible. Good clean blood is the source of all life to the cells and nerves and brain. To clean the blood of toxins will enable this dynamic fluid of our bodies to provide our organs with proper nourishment. Spirulina plankton is the most natural way to provide strength and power and purify and fast at the same time.

Spirulina plankton is several times more potent than wheat sprouts or wheat grass in providing the proteins and essential amino acids to complete nutrition. Although grass or wheat is all that a horse or cow requires to become fit and healthy as a range animal, the Spirulina is known to be five times more nutritious for humans.

The following instruction should be followed carefully because failure to do it right may end up with your enema bag or syringe tube blocked up with clotted plankton.

Take the following ingredients:

- 2 teaspoons Dr. Hills Spirulina green spray-dried plankton
- 2 teaspoons of glycerine U.S.P.
- ½ cup cold distilled water
- 1 blender full of distilled water
- 1 enema bag or syringe

METHOD

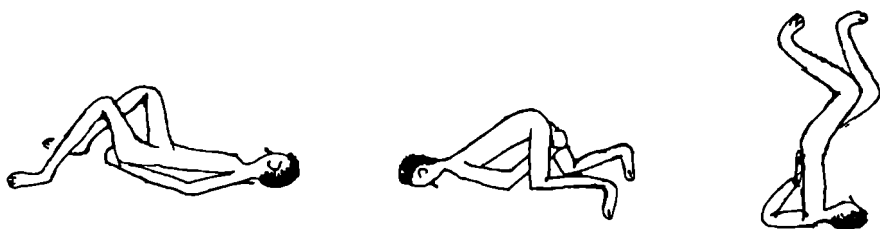
Heat the water to just above blood temperature. Mix 2 teaspoons of plankton green powder with ½ cup of cold water till it becomes a smooth paste and then add 2 teaspoons of glycerine and stir up well together. Add the plankton paste to ½ full blender and operate on "stir". (Liquid can be beaten with egg whisk if blender is not working or not available.) When thoroughly mixed up, add remaining distilled water as the mixture is stirring in the blender to fill capacity of blender. Fill enema bag with the result and use quickly before temperature cools below blood warmth.

By washing out the lower and upper bowel with a full injection of water mixed with Dr. Hills Spirulina plankton chlorophylls, a proper peristaltic downward action of the whole alimentary canal is performed. By absorption of the plankton in the cell wall mucus and the loosening and diluting of the bowel contents, the whole of the abdominal intestine is cleaned out and purified. It is important to use the following method to ensure that at least 1 quart (and preferably with practice 2 quarts) of the plankton water penetrates the folds and corners of the sigmoid colon, the ascending colon and the transverse colon into the descending colon. It will take 2 quarts to fill the entire colon but

beginners should start with a full bag and see how much they can tolerate. In flushing the entire colon the mixture is gradually injected a little at a time while lying on the right side and turning over to the other side while injection takes place. Massage of the abdomen helps to send the mixture through to the upper colon. Another method is to apply the enema syringe in a kneeling posture with one's head on the level of the ground in front of the knees. Gently rock the body sideways while the mixture is undergoing injection into the anus. Before applying the syringe into the anus smear some vaseline or oil on the surface to help a smooth entry into the bowel. Most syringes extend 3 or 4 inches so as to pass the sphincter muscle in the rectal opening. If during the injection of the mixture the person feels a reaction or discomfort shut off the syringe to cut off the pressure until it has been absorbed and then let out some more mixture into the anus. If the person feels that retention of the mixture is impossible then the enema may be ejected into the toilet bowl immediately and the process started again. With practice the bowel will become adapted to injection of the mixture and eventually with more frequent use of the enema the bowel will be able to retain a full enema of 2 quarts.

After injecting the full mixture it should be kept inside the bowel for 5 to 10 minutes while massaging the intestine through the abdomen to loosen and soften any hardened fecal matter and to circulate the mixture throughout the intestinal tract. Those familiar with the shoulder stand, whereby the arms hold up the buttocks in the air, will gain increased circulation of the plankton enema into the higher intestine if they include this posture after injecting the enema in the rectum. The addition of glycerine not only helps to emulsify the mixture in the blender but also helps to quickly soften the impacted feces and lubricate the walls

of the intestine.



Begin the enema the first night of the fast. Even if you are not fasting and are using the Dr. Hills Spirulina plankton as a new dietary supplement, the enema will help the intestinal flora to adapt to the high percentage of chlorophyll and other cleansing pigments in the new food very quickly. Apply the enema three nights in succession, then every other night for three nights while drinking a larger amount of fluids than usual. If you like water drink it frequently when thirsty, then juice may be taken at meal times with the Spirulina plankton, but if you intend to fast longer than a week you should try to drink vegetable and fruit juices often to flush out the system. Once the fast is over an enema once a month is sufficient to maintain health. However if there is a record of much ill health, flatulence or constipation an enema once a week may be in order until perfect health has been restored.

Do not use any other but the Dr. Hills Spirulina plankton *spray dried* because freeze dried or drum dried does not go into suspension and is organically cooked dead matter. The enzyme in Dr. Hills Spirulina plankton is still alive and it will bond with the intestinal mucus leaving you feeling very clean and fresh in mind and body. For more details on enema and health practices associated with

blood purification and breathing you may refer to the book of my colleague:

May You Live in Health

by Dr. Aaron Friedell, M.D. available c/o University of the Trees Press.

FOR JOGGERS AND ATHLETES, SPIRULINA AND RAW VEGETABLE JUICES

One of the best ways to use the wonderful nutritional supplement called Spirulina plankton is to mix it with raw uncooked vegetable juice. In this modern society we live a paradoxical life of dietary inadequacy in a land of plenty. We eat great amounts of food but our cells are starved of the correct nutrients.

Natural fruits and vegetables supply the body with missing vitamins. Spirulina is one of those remarkable vegetables which contains all of the magic minerals, cell salts, amino acids, enzymes, etc., which are vital. To make Spirulina into a juice or to use it in foods is a skill we must learn to become perfectly healthy.

Raw vegetables should be fresh and free of preservatives and chemicals in order to capture the life force. Combined with the super vegetable plankton they can convert into body energy in less than fifteen minutes after drinking them. As a source of quick energy, Spirulina plankton in a vegetable juice makes available nourishment for the body's cells and tissues in less than ten minutes after ingestion. Can you imagine how important this is to joggers who not only need this energy for stamina but who need to replace many of the minerals and vitamins leached out by one of the body's most efficient elimination systems—

the sweating skin?

Depending on the person's activity two to six pints of juices can be ingested daily without any problems. The body recovers its losses through elimination by extracting the nourishment from juices entering the mouth. These juices give our bodies the power to cleanse and heal, not only because we need their valuable renewing minerals, vitamins and enzymes, but because they give us instant power and energy.

Many diseases can be gradually eliminated through raw juice therapy but here we are concerned with detoxifying the blood quickly so that glands can function at their optimum level. To get blood to circulate and flow efficiently to the muscles and joints we must release the immediate powers of Spirulina into the body. The addition of vegetable juice replaces the natural juice of the original vegetable plus the enzymes in the reconstituted juice of the Spirulina which is blended with it. Thus you begin to start feeling better very quickly and the toxic wastes and impurities in the blood are released into the body's eliminative systems. This re-establishes your natural energy contact with the subtle body which is directly linked to the Cosmos.

Only the blockages in the physical and subtle body* prevent your receiving unlimited energy from the total environment.

Of course if you continue to eat ice cream, candy and rich creamy pastries and eat food which comes out of cans

* See *Subtle-Body Healing* by Victor Beasley for further discussion. Available from the publishers.

then the juice therapy merely keeps up with your abuse of your body. The carbonated drinks mixed with sugar and glucose pile up their effects in the body. Carbon dioxide mixed with sugar goes immediately into the brain but causes a dependence and may later result in gas and distention of the intestines. Bloating of the eliminative organs through fermentation prevents the easy assimilation of raw juices and our body ages faster. Tensions pile deep into the operation of our organs and nerves and raw vegetable juices enable us to release them by releasing the toxic wastes through the skin. The sweat of a healthy person is not unpleasant but bad breath and B.O. are indicators that we need to excrete toxic material and replace it with nourishing food for the cells.

The new energy must force out the toxins accumulated in the body tissues and health and natural stamina can only come when these polluted conditions are no longer blocking the cosmic flow.

Following is a list of vegetable juices which can be mixed with a teaspoon of Spirulina plankton to make a source of quick energy. There is a difference between taking capsules directly into the stomach and mixing Spirulina with fresh vegetable juice. Four 500 mg. tablets of Spirulina powder equal a level teaspoon, but unless we savor each mouthful and mix the juice with our saliva the full benefit is not received. Although 500 mg. tablets are a convenient way of getting the benefits of Spirulina plankton they are not the same thing as a raw juice that is rolled around in the mouth and then swallowed, which gives us instant assimilation through the addition of digestive enzymes from our own mouth. For joggers and athletes this is important.

These juices make excellent mediums in which the super food values of Spirulina can be reconstituted as a fresh juice for immediate assimilation:

Raw vegetable juice	Combines well with:
Alfalfa	carrot
Asparagus	beet, cucumber, carrot
Beet (small wine glass only)	carrot, lemon
Carrot	general mixer
Celery	carrot
Cucumber	spinach, carrot, lettuce
Dandelion	carrot
Endive	carrot, celery
Lettuce	cucumber
Parsley	carrot, endive
Pepper (green)	carrot, spinach
Potato	carrot, celery
Spinach	carrot
Tomato (no sugar or starch added)	carrot, celery
Turnip	watercress, carrot, spinach

Other liquids and juices can be taken throughout the day but to get quick energy at least a pint a day should be taken.

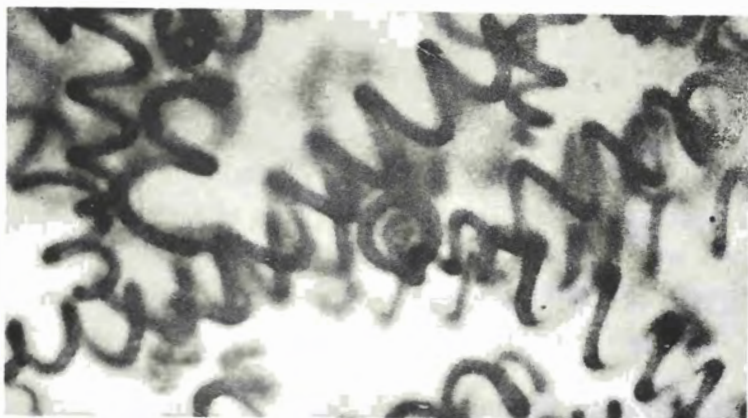
To use the raw juices and Spirulina as a cleanser for fasting the intake of these juices can be alternated with fruit juices freshly made. Nature tells you automatically when you have had enough, so do not overdo it by force feeding yourself. You drink what feels comfortable to you. Six pints of juices during a day mixed with a teaspoon of Spirulina can easily be taken on the normal fasting diet, but if you are an active person and jog a lot you can increase the quantity until you have completely resupplied the body with revitalizing fresh enzymes and natural elements.

OTHER USES FOR SPIRULINA

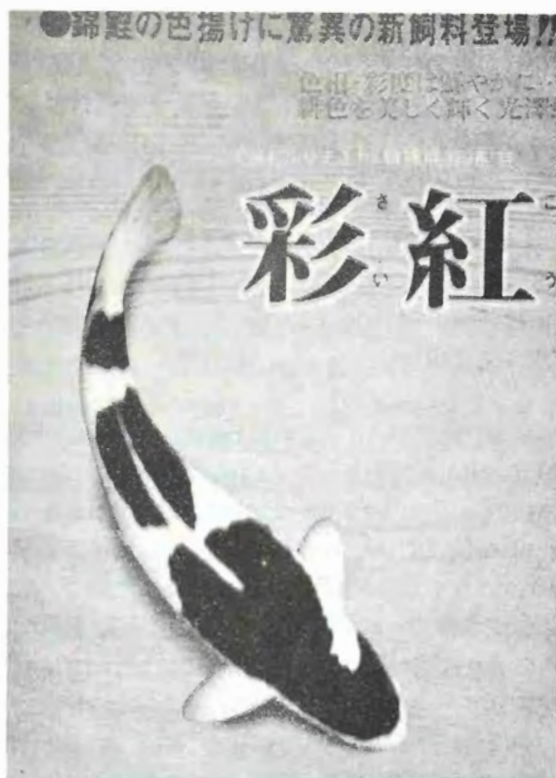
In Mexico the natives around the lake use the vegetable plankton as a hair conditioner. The chlorophyll and vegetable protein reconstructs tired hair and someone there even markets a special brand of hair treatment. In France the Spirulina is made into a paste and rubbed onto the skin and face as a skin conditioner. Of course Spirulina used in the intestine to create good elimination also makes the skin smooth and young, but some people who are interested in French beauty products have made it into an external application. In Japan a manufacturer claims that Spirulina actually helps to grow new hair through using it as nutrition.



Algae cakes drying in the sun in the Republic of Chad.



Cell structure of Spirulina as seen through a microscope, showing spiral nature of the vegetable plankton.



Japanese packaging of a Spirulina product for the fast growing of competition fish—used by fish fanciers to improve color and vigor.

HOW TO GET SPIRULINA

Spirulina vegetable plankton is grown in laboratories under special conditions and is not available in commercial quantities anywhere in this country. It is grown in Siam, Japan and Mexico, where it grows naturally in Lake Texcoco. Dr. Hills Spirulina Plankton is 100% pure without additives of any kind.

It is available from several sources in different packagings as follows: 1 lb., 3 lbs., and 5 lbs. containers for use as a general purpose powder for mixing in juices, smoothies, etc. Available in many formulations from:

**AQUACULTURE NUTRITION PRODUCTS CO.
LIGHT FORCE SPIRULINA COMPANY
AQUACULTURA NUTRICIONAL DE MEXICO S.A.**

Spirulina powder is put up in bottles of 100 and 200 tablets, 500 mgs. per tablet, and can be obtained mixed with various other nutritional ingredients to form tablets for different nutritional purposes. More details of these can be provided by the author.

Not only is Spirulina highly beneficial as a nutritional supplement by itself as a whole food, but it also combines well with other vitamins that contribute to produce specific nutritional effects in the metabolism of the body, without resorting to synthetic vitamin combinations. For example, Spirulina Plankton together with Niacin B3, Pyridoxine B6 and Calcium Gluconate helps the metabolism and circulation, activates hormone production of adrenalin and insulin and strengthens the nervous system and muscle impulses. B15, or pangamic acid, was used by athletes and tested for some years by Russian and European researchers, but there is actually no need to buy the special so-called

miracle B15 because eating calcium gluconate, niacin and Spirulina all at the same time activates the cellular metabolite responsible for the biological action associated with vitamin B15. So the vitamin is naturally created in our stomach without our having to purchase any special synthetic preparation. Joggers, athletes and tired people who need extra energy will be happy to discover that they no longer need to rely on synthetics.

Another beneficial combination is Spirulina with Ginseng, Bee Pollen and Papaya enzyme. Health and nutrition enthusiasts are already familiar with the properties of Ginseng, Bee Pollen and Papaya, and will appreciate their value as special supplements to Spirulina's rich, natural source of minerals, vitamins and amino acids.

Cultured Spirulina plankton and cultured brewer's yeast balance each other. Spirulina is grown in large tanks by photosynthesis from the sun's light and primary grown yeast is produced without light in sealed fermentation conditions which exclude all contaminants. The pure culture of the microalgae Spirulina and the chemically pure enzymes in brewer's yeast not only both contain the world's highest source of protein and B vitamins, but also biotin, choline, lecithin, fiber, inositol and balanced minerals such as iron, copper, zinc, calcium, magnesium and phosphorus, and include all essential amino acids.

A combination of Spirulina Plankton and ground and powdered marine shells yields a wide spectrum of sea minerals and vitamin B12 together with the other constituents of Spirulina set out elsewhere in this book. This entirely natural product contains no additives of any kind and consists of 50% pure Spirulina plankton, the world's highest source of natural nutrition. The remaining 50%

consists of a seafood product made of ground and powdered marine shells which, on analysis, yield amino acids, essential trace elements and minerals from the shell. The minerals from the sea, including calcium in the shell, are formed biologically the same as in the *Spirulina* and are balanced the same as in the human body. These ocean minerals are balanced and naturally chelated by the shellfish as it grows and thereby afford more natural efficient assimilation into the body. Heavy metals such as cadmium and mercury are not detectable in this sea product.

Spirulina is the world's highest natural source of vitamin B12 containing 255 mcg. per 100 grams as against 40 mcg. for *Chlorella*, 14 mcg. in fermented soy foods such as tempeh, 27 mcg. in seaweed, 2 mcg. in eggs, 1 mcg. in cottage cheese, and only 0.42 mcg. in 100 grams of milk. Even liver, the highest source of B12 in the animal world, is only 80 mcg. compared to *Spirulina's* 255 mcg.

Comfrey root and its leaves (*Symphytum*) is referred to by herbalists in glowing terms for its soothing effects on swellings and as a health tonic for the lungs.

Even medical science is taking a second look at this old time herbal drink which has been used for centuries as a demulcent, astringent and effective cell life proliferant. It was used for many years in poultices and fomentations to assist the healing process of cuts. It received the common name of "knit-bone" because its fomentations would ease the pain of bruises and sprains. Now taken internally for health and nutrition purposes, Comfrey root can be prescribed as a demulcent to soothe the mucous membranes of the stomach and intestines. Comfrey is also taken regularly for the healthy nutrition of the kidneys, bladder and uterus.

The herb contains *allantoin*, an active metabolic agent of cell proliferation and for gastric health. Farmers have long known that Comfrey prevents and cures intestinal and digestive conditions in pigs, cattle and horses. Present FDA laws prevent nutritional consultants from prescribing it for humans as a cure for intestinal disorders so they are not allowed to say whether it performs equally well for humans as it does for livestock. But for a healthy intestine it is highly esteemed by all nutritionists as the most important herbal supplement which contains almost 1 percent of *allantoin*. Combining the root of this herb with the most active biological food known to man—Spirulina Plankton—makes a most potent supplement. Its name Symphytum means “to join things together” or “to grow together”. It was given because of its power to promote growth of cells when applied fresh to wounds and now we use it to enhance the health giving properties of Spirulina.

Comfrey has been used as a blood purifier for centuries and the leaves were eaten in Europe as we eat spinach or chard. Everyone knows that pure blood builds healthy tissues and as a health supplement it is worth eating comfrey even if it is only to keep the blood stream pure and improve circulation. An addition of 10% Pepsin to this Comfrey/Spirulina combination helps the gastric juices to break the amino acid linkages in the Spirulina, which contains over eighteen amino acids, ten more than the eight essential amino acids needed by the body from external sources.

SUPERSENSONIC POTENTIZATION

It would occupy too much space to explain the full process researched in my other books which I call “Supersenssonics” and later developed at the Research Institute

for Supersensonic Healing Energies (RISHE). Some brief comments about the method of potentizing a product by homeopathic principles is necessary.

Anyone familiar with the Homeopathic potentization of cell salts sold in every health food store will be able to understand the principles behind the activation of the amino acids and trace elements in Spirulina Plankton. Anyone who uses a pendulum or does kinesiology will be able to test out the difference between ordinary foods and those which have been vitalized. The whole concept of blessing the food we eat is founded on the idea that some energy is somehow imparted to the blessed food. To claim that Spirulina Plankton is specially blessed may raise some eyebrows, so we set out below some highlights of the Supersensonic process.

Spirulina Plankton is not like any other vegetable algae or vitamin and can be made even more superior to any other protein when the plankton is specially potentized by a new homeopathic method known as *Supersensonic geometrics*. Briefly, it means taking the geometric patterns of a crystal, a salt or a cell and using orientated magnetic fields to boost its potential life energies in the higher spectrum of energies concerned with the spiritual, mental and subtle levels of our consciousness. It is no less than applying the powers of the mind to do what Christ taught when he amplified the barrenness of the fruitless fig tree and withered it. Supersensonics is doing the same thing to the plankton in reverse, enhancing its fruitful properties for good.

This new science of Supersensonics is not taught in colleges nor approved by any authorities except nature herself. Who will believe that you can take a 100% natural product and energize the trace elements through amplifying the effects of consciousness on the subtle forces of the life

energies? If you wish to know more about this new science you can find it adequately described in a series of practical books as follows:

Supersensomics , by Christopher Hills, Ph.D., D. Sc.	\$15.00
Subtle-Body Healing , by Victor Beasley, Ph.D.	\$5.95
Your Electro-Vibratory Body , by Victor Beasley, Ph.D.	\$9.95
Wholistic Health and Living Yoga , by Malcolm Strutt, M.A.	\$9.95
Energy, Matter and Form , by Allen, Bearne and Smith	\$9.95
Nuclear Evolution: Discovery of the Rainbow Body , by Christopher Hills	\$12.95
Rise of the Phoenix , by Christopher Hills	\$14.95
Secrets of the Life Force , by Christopher Hills	\$5.95

All of the above books are available for deeper study from University of the Trees Press, P.O. Box 644, Boulder Creek, California 95006. Add \$1.00 per book postage and handling.

A user of potentized Spirulina is entitled to an explanation of why it is different from other Spirulina and what claims are being made for the product. The laws for mislabelling a food or drug product are very strict because of so much fraud and misrepresentation among big and small firms alike. Adulteration with preservatives and fillers and binding agents in tablets is common throughout the trade. *You should make sure all Spirulina products have no fillers, binders or additives of any kind and are 100% pure algae as grown by sunlight and Mother Nature.*

In addition to the advantages of the wonderful Spirulina vegetable plankton in formulating nutritional products, the following procedures of Spirulina potentization should be followed:

The plankton is analyzed Supersenssonically to discover the homeopathic effect of its total health-giving properties with the special instrumentation developed by Dr. Hills and his colleagues for energizing certain selected amino acids and trace elements which give the Spirulina a natural advantage over other forms of nutrition. These 12 properties combined in Spirulina are picked out and potentized by charging them with the Supersenssonic Homeopathic Energizer to boost the pronounced buzz of energy which comes by combining and mixing the nutrients of natural Spirulina in the stomach. Supersenssonic instruments charge the Spirulina trace elements with subtle radiational life energies which work on the principle of mind over matter, commonly referred to as psychotronic instruments. There is no physical way of testing the enhanced products except by the muscle-testing methods of kinesiology or by dowsing with a pendulum over one product to compare it with another.

Radionic and Supersenssonic energies are defined as nonphysical thought fields which are concentrated and focussed by psychotronic instruments developed by the Research Institute for Supersenssonic Healing Energies (RISHE) for experimental purposes in the potentization of trace elements and homeopathic cell salts for nutritional health purposes. In practice the potentization can be applied Supersenssonically to any substance in any quantities by anyone with a full conscious understanding of the new science of Supersenssonics. Students can enroll through the University of the Trees for courses in these new and original techniques of transmitting life force. Because it is a completely new breakthrough in consciousness research at the frontiers of human knowledge, these methods are not known or accepted by traditional universities or by the official medical authorities who at present would oppose their use in healing. Several individual doctors who have tried to use these methods in the past for healing have been harassed and even imprisoned for practicing radionics and labelled as frauds. It is unfortunate that in every new field or development frauds do jump on the bandwagon and imitate without real knowledge. It is therefore necessary to

make the disclaimer, so that anyone who buys Spirulina Plankton is made aware and not subjected to false claims or fraudulent mislabelling. A sample disclaimer follows:

The FDA and California Law prohibit any person from prescribing or advertising any product or system for diagnosis or treating ailments or medical diseases or recommending anything for the mental or physical condition of any person, but does not prohibit the provision of nutritional advice for proper nutrition, dietary supplements and the role of food and food ingredients. No medical claims are implied or warranted in the description or labeling of Spirulina health products and prescription is solely for good health practice and proper nutritional advice.

The only real test is in how you feel and the level of life-force you manifest after using the product. All claims for any nutritional product must inevitably be confirmed by the user's own special metabolism. The author is interested in receiving any reports on the effects of these products for publication of research in the near future and spreading these Supersensonic concepts to a wider public. The University of the Trees is licensed by the State of California to issue authorized degrees in the fields of consciousness research and wholistic studies and is particularly interested in publishing works on the nutritional effects on consciousness.

Persons wishing to know more about consciousness research studies can send \$1.00 plus 50¢ postage for the College Catalog of courses. All students are encouraged to develop their own business as explained in the idea of common ownership, the sharing of wealth and abundance as God-given products of consciousness.

BACKGROUND

There is much in this booklet I cannot write about *Spirulina* which I have acquired over 25 years of contact with those who are doing things all over the world to feed the hungry or develop new uses and products which can use the remarkable properties of *Spirulina* for the good of everyone. As President of the Microalgae International Union I have met many scientists who think they know a lot about algae and some who are doing projects but have no idea as to what has already been done. I myself have done projects in the West Indies and India which eclipse in size anything being done presently in America to produce algae.

Mexico, Kenya and Chad all have lakes in which *Spirulina* grows naturally without any assistance from man, and the Mexicans have done remarkable work in assisting its growth in large quantities. In 1968 the production was small. My colleague and best friend, Professor Hiroshi Nakamura of Tokyo, visited Mexico that year and we together realized that production could be increased there to large commercial quantities. But to the credit of certain Mexican individuals who pioneered the production of *Spirulina* as a by-product of the production of soda ash and caustic soda from the concentrated brine of the lake, the crop is now large enough to launch a major demonstration. Until recently the Japanese technology on algae was supreme and probably still is, but the natural growing conditions of Mexico make it a more favorable environment. The reason is this: Japan has little land and no natural alkaline condition for *Spirulina*, and Lake Chad and Kenya are far away from a sophisticated center of engineering, technical personnel, finance and industrial laboratories, etc. Mexico City, whatever Americans might believe, has the largest population of educated Latin

Americans in the world. Mexico has a population more than that of Great Britain and Ireland put together and a European-educated elite which makes the Mexican nation a totally different entity to what most Americans believe. Sixty percent of Mexican territory—covering the states of Texas, New Mexico, Arizona, California, Nevada, Utah and parts of Colorado and Wyoming—was taken over by the US in the 1840s. The annexation of Texas in 1845, which caused the Mexican War in 1846, was opposed by a congressman named Abraham Lincoln who called it aggression of Mexican territory. Most Mexicans still resent this, just as the Palestinians resent the annexation of Jerusalem by Israeli troops, and just as the Afghans and Czechoslovaks will resent the invasions of the Russians for years to come. This presents a cultural barrier which modern Americans cannot understand. The truth is that Mexico City is a modern city of 15 million people, which makes it the largest city in the world, even larger than London, Tokyo or New York. As such it has the best and the worst of all worlds. But to ignore the best because of the worst is the American problem. We just cannot believe that the Japanese who copy everything America makes or the wetbacks who flock across the border can have anything going for them. Those stereotype images are America's problem in others areas of foreign policy, but in the production of algae it means that no one thinks they can learn anything from Japan or Mexico. The facts are totally different. The production and consumption of *Spirulina* in those countries far outstrips anything the USA is doing or even thinking of doing.

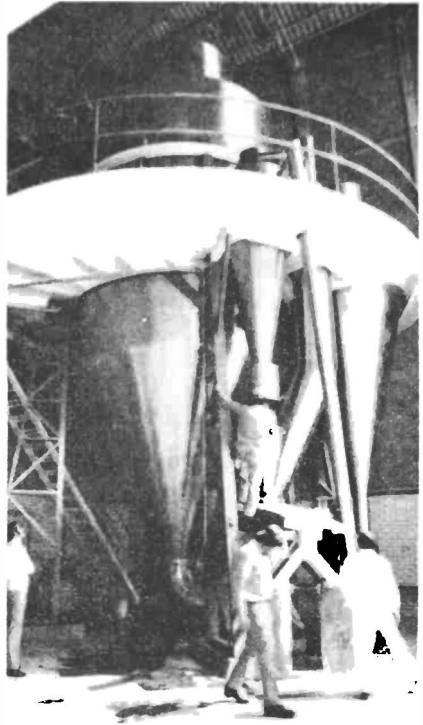
Perhaps I should declare that my personal interest in *Spirulina* over the last 25 years has led to various associations and production projects. Apart from having a 10 percent interest in several American companies which have failed to produce *Spirulina* over the last few years the writer is a part-owner with other Mexicans, (Mexico now has a production of *Spirulina* of 600 tons per year),

Commercial production of Spirulina plankton underway in Mexico.



A view of the extensive growing and harvesting ponds showing the Spirulina in its final stages of production.

For comparison 8 grams of Spirulina contains approximately the same vitamins, proteins and amino acids as 100 grams of Tofu. Spirulina is probably the world's highest source of B12.



A picture of the spray-drying equipment, showing the Spirulina plankton taken straight from the ponds, harvested fresh and packed into drums.

The capital investment in these large-scale ponds and equipment is enormous. AQUACULTURA NUTRICIONAL DE MEXICO S.A. is seeking government loans of \$48 million to grow Spirulina in all other alkaline lakes of Mexico. But anyone with high school botany experience can grow Spirulina in tanks or bottles in their home once they have the correct mix of salinity, alkaline medium, nutrients and temperature. Persons interested in investing in algae production may engage Dr. Hills as consultant. He does not accept direct telephone calls but will answer mail.

of a company dedicated to marketing Spirulina throughout the world. Since 1970 production has been steadily increased. Aquacultura Nutricional de Mexico S.A. is represented exclusively by the author who also represents producers in Japan, but the Mexican company is willing to supply and make up other products under private labels for anyone who wants to buy a large tonnage. But we will not talk to people who don't have the courage to buy some tonnage because so much hot air passes through the minds and mouths of speculators who do not risk a penny to find out what took us 25 years to find out—that Spirulina is truly the Green Gold of nutrition.

ALGAE BANK

It has been a wild idea in the back of my mind now for about 10 years that although gold is the best hedge against depreciating currencies, you cannot eat gold, so if there is ever a survival problem where people need to get concentrated nutritious survival food in times of emergency the real currency will become a one pound packet of Spirulina plankton. To sell all your gold and buy Spirulina plankton and put your faith in algae food instead of gold might not impress the ultimate materialist as a wise move, because he thinks that in times of need anyone will sell food for gold. But will you sell a survival store of plankton that will keep you in perfect health for several years to the first person that comes in your door? You would certainly not accept worthless paper dollars which could in a hyper-inflation take as many as 10,000 or even 50,000 to buy a newspaper. As ridiculous as this sounds there are certain fatal basic assumptions about money that always catch the unwary. Under emergency powers the cash in your bank account can be frozen and, even now, under federal law it is subject to 30 days of withdrawal notice at any time. Most people do not know this about their current accounts

because they do not read fine print and this clause is not exercised *at the moment* by banks. But come a presidential decree and overnight you could not withdraw more than customary amounts from your savings or current checking accounts. Even gold in safe deposit boxes is not safe from the Emergency Banking Regulation No. 1. Hence I realized that a currency backed by algae instead of gold is more safe because people would only accept gold for it if they did not need the algae to eat for themselves. Ridiculous, you say, it cannot happen here in America! But the facts are that you don't know what the American Government can do under emergency powers which will be administered by government bureaucracies if and when these laws are ever invoked overnight.

I have not started such a bank which would issue warehouse receipts for stored algae rather than money because only the government is allowed to create money and it is therefore illegal to print money as a private citizen. But algae certificates would soon achieve the value of green gold certificates in an emergency because ordinary money and goods and prices will be rationed and paralyzed by emergency government controls. If you own a warehouse receipt it means that product is already your property and you will beat the system of price controls when the economy will go bankrupt. I have written about these ideas in *The Golden Egg** so we will not labor the reader here, but merely say there are other uses for Spirulina as a store of value far superior to gold. It would not be feasible to start an algae bank unless one had at least 5,000 people interested in it as the survival food of the future. The fact that astronauts can use it on long trips makes this concentrated source of nutrition not only earth's finest food source but a "Spacefood".

* Published by the University of the Trees Press, 1979.

Recently I have investigated the mixing of Spirulina as a supplement to horse feed for the enhancing of performance. Horse racing and competition events have brought Spirulina to the forefront of animal nutrition. A balance between the carbohydrate and proteins of horse feed can be achieved with the addition of 35 percent Spirulina which gives any horse lover a much friskier horse even after a few days. Additionally I have experimented using Spirulina in breakfast foods and cereals such as granola. I never liked the granola I found in health food stores until I tasted Spirulina granola. Now I find mixing it with natural sweeteners and carbohydrates makes this breakfast food into the highest protein cereal supplement on the market.



Picture shows Prof. Hiroshi Nakamura and Toru Matsui, an 85 year-old philosopher who has lived on Chlorella and Spirulina algae for 15 years. Spirulina contains enough vitamin C for living; all other vitamins in Spirulina are in excess of human needs. They are eating algae and getting younger.

SUMMARY

Many people who suffer from low blood sugar get symptoms of headache, exhaustion, etc., so when they fast and release toxins into the bloodstream through a cleansing diet of Spirulina, they think they need to eat to keep up their energy. But this is not so if the rules of fasting are followed properly. Initially most people begin to feel sickly during the second day of the fast because they don't follow instructions properly. If you do feel headaches and upset it can be due to:

1. toxins releasing from cells into the blood
2. dilation of blood vessels in brain due to serotonin and hormonal release
3. not enough liquids, juices or water. If juices are not desirable then drink copious amounts of water to avoid constipation. Bran can be mixed with the Spirulina so as to get it to move through the bowel. Making a broth, straining the liquid and mixing a spoonful of the Spirulina before drinking is another way of preventing a loss of energy on the second or third day.

When a meat eater is coming off the fast he or she should not eat meat right away because it returns old flora to the intestine after you have taken the trouble to encourage new intestinal flora. If you feel the need later you can eat an alternative protein diet, but with Spirulina the protein is four times that of beef. To produce new flora we can eat acidophilous yogurt before eight o'clock in the morning because this establishes living bacteria in the intestine. To eat yogurt during the day and evening, however, is mucous-producing. Yogurt mixes well with Spirulina and helps to rebuild a healthy flora.

A most important factor in fasting is your motive. Most people fast to lose weight or they feel their digestion is off and think a fast would help. But some people fast for

spiritual reasons and achieve a purification of the spirit as well as the body. The reasons for spiritual fasting are not always given clearly. The clearing of the mental faculties enables us to see life around us more clearly when we are fasting as a form of worship. (Worship means making oneself worthy). We can change our motive and way of eating by giving thanks and programming our consciousness into the food like a blessing. By thinking of food as a form of light and realizing that Spirulina is imprisoned light, we are able to trigger other feelings and emotions which release brain chemicals.

Research is at present underway at the University of the Trees to potentize the Spirulina with the Supersensonic vibration of the major brain peptides and endorphins which change our states of consciousness. In the same way that we can activate the enzymes and amino acids in Spirulina by homeopathic potentization, the Research Institute for Supersensonic Healing Energies (RISHE) is looking into the programming of the geometrics of the following brain chemicals which contain many of the amino acids in Spirulina.

- | | |
|--------------|----------------|
| ● ENKEPHALIN | ● DOPAMINE |
| ● MELATONIN | ● NOR EPINEPH- |
| ● GABA | RINE |
| ● OXYTOCIN | ● ACTH |
| ● SEROTONIN | ● LUTENIZING |
| ● ENDORPHIN | HORMONE |

This will be the subject of a separate book after the results have been thoroughly tested out with students and experiments. In the meantime the future of the Spirulina plankton as a source of energy and chlorophyll is as unlimited as “manna from Heaven”.

SUMMARY OF REPORT FROM DAINIPPON INK & CHEMICALS, INC. JAPAN

LINABLUE-A—A Natural Blue Colorant from Spirulina plankton

Linablue-A is a new and unique water-soluble natural colorant in brilliant blue which has been extracted from the blue-green plankton *Spirulina*. Dainippon has paid special attention to Phycocyanin, a colorant peculiar to *Spirulina*, and after elaborate long-term studies, has commercialized it as Linablue-A, a brilliant blue natural colorant for foods. It is a pale blue powder with a faint sweetness but with no odor, and when dissolved into water, it offers a brilliant blue color with faint reddish fluorescence. Phycocyanin has a special chemical structure which bonds naturally with proteins to create linkage with peptides.

USAGE:

Many tests for toxicity of this natural product have been done on duck embryos and mice and unlike many synthetic food colorings Linablue-A was found to be completely safe when administered to laboratory animals. The manufacturer suggests it for a variety of uses: iced confectionery (including ice cream), confections, dairy products, powdered reconstituted soft drinks.

PHARMACOLOGICAL STUDIES ON SPIRULINA

**(Department of Pharmacology, Meiji College
of Pharmacy, Japan)**

A team of researchers in Japan recently concluded extensive experimentation into possible hazards or side-effects from ingesting *Spirulina* plankton and to discover the maximum dosage. Their conclusions:

In studies using laboratory animals they found no adverse effects whatsoever upon the central nervous system, upon brain waves, blood pressure or heart rate. Some movement in the intestines could be noted, but no diuretic effect was evident. In summary, the physiological activities and toxicity of Spirulina in experimental animals are extremely low.

In a further study on the effect of Spirulina on the reproductive functions and offsprings of rats, three doctors at the Animal Reproduction Research Laboratory in Japan found no disturbances attributable to Spirulina and concluded that Spirulina has no adverse effect upon their reproductive functions. In fact, the doctors added, it was confirmed that Spirulina can be claimed to be an outstanding foodstuff, as demonstrated by the rats' growth during experimentation.

These voluminous scientific reports on the clinical efficiency of Spirulina would occupy a whole book with tables and procedures which only would interest researchers interested in duplicating the experiments. The fact that very little work with the medicinal value of algae and plankton has been done in the U.S.A. may be due to the tendency of Americans to lean on drugs which alleviate symptoms rather than achieve cures.

APPENDIX

The following personal accounts are by people who have undertaken fasts and wish to share their experiences of Spirulina. The first is by Nancy London, a former manager of the "Way of Life" herb store in Santa Cruz, California.

"Fasting with Spirulina Plankton certainly is different from any other fast I have ever done. I've tried them all. Lemon juice and water fasts. Wheat grass juice fasts. Potassium broth fasts. They all had their strong points but each one left me tired and in need of a lot of rest while my body cleansed itself of toxins during and after the fast. I work hard now—a busy schedule day that leaves little room for prolonged periods of inactivity. So I've kept away from fasting because I need to stay active.

"When I heard about the high protein content of Spirulina (five assimilable grams per tablespoon!), the vitamins (A, B complex, etc.) and the minerals, plus how it lines the intestinal walls and aids elimination, plus how the high chlorophyll content helps purify the blood of toxins, a bell went off in my head. This might be the perfect substance to fast on. The body could cleanse itself while remaining energetic and well nourished.

"I went on a Spirulina fast, consuming nothing but Spirulina Plankton powder and juice or broth for one month. I hadn't planned on fasting that long but it felt so good I just kept on going. Here's what I found out for myself:

- I experienced no hunger because of the incredibly high nutritional content.
- I stayed active and energetic with an average of five hours sleep a night.

- I lost weight naturally without having to starve myself.
- I eliminated almost daily for one month, leaving me feeling cleansed and purified.
- My skin and hair became radiant and healthy.
- My concentration and meditation improved with the elimination of toxins from my bloodstream.

“Now my friends are trying it for themselves. They’re looking younger and slimmer, like they’ve found the fountain of youth! It’s an exciting discovery and we’re glad to be able to share this amazing health secret with you too.”

An account from Pam Osborn:

“My first fast with algae went as long as 12 days, consuming only algae with fruit and vegetable juices. It was a far more pleasant experience than my previous fasting experiences because there were no hunger pangs and my energy level was higher. Because of this I was able to fast for twice as long as I had ever fasted before.

“At a later date I fasted for 11 days on the *potentized* algae and, amazingly, found this to be at least twice as energizing as the regular, unpotentized algae. In fact, my energy level was higher than normal times when I ate food! I was up at 5:20 a.m. to jog and I worked late into the night, never once interrupting my busy schedule to rest. My mind was clear as a bell and I’ve never felt better, ever. For this particular fast I decided to try something I thought would be an ideal combination with algae to facilitate the toxic elimination process. With my algae drinks I mixed in raw, unprocessed bran. Bran, though completely indigestible, provides the bulk to keep things moving through the intestine quickly, sweeping up wastes dumped there by the

cleansing body. I found that, unlike previous fasts, I never needed to take enemas, because I never reached the point where I was feeling headaches or sickly which is due to toxins polluting the intestine and then affecting the bloodstream.

“I have come to conclude through experimenting with many foods that an algae diet with the addition of bran for roughage is the best diet I could possibly take for good health. Unfortunately, I am at this point unable to give up the taste pleasure of ordinary foods, but one day I plan to convert entirely to an algae diet for my sustenance.”

An account from Sibyl Greene:

“For 30 days last summer I lived on algae and fruit juice. This provided all the nutrition I needed without any discomfort so I felt light and satisfied. Previous fasts had, after 10 days or so, left me feeling weak and hungry, whereas the algae never did. Only once during the whole month did my energy drop off a bit and on consulting with Dr. Hills, I ate some extra algae and felt my energy return almost immediately. The effect of the algae fast was a cleansing and detoxification so that I felt more at peace and in harmony with the natural world, clearer with less tension. Normally, if I eat some sugar or coffee, my body responds chemically as if I had put an electric charge through my system. Then after this stimulus/response is a recovery period of fatigue and the resulting stress and tension of waiting for the next cycle. So also on the average American diet the human system is constantly in some state of stress. Once I had been on the algae for a week or two, my system began to learn that I was no longer to put it through this ordeal several times a day but rather that what I was putting in was in tune with its own rhythms and movements and the experience would be nurturing and enriching the cells. Eventually my nervous system became

more sensitive and could 'read' the messages in and around me more directly. My attention was sharper and concentration increased. My mucous membranes cleared so I had no congestion or sluggish feeling. Even in the early morning after little sleep I woke up lively and feeling light without the veil of somnolence that often follows overeating or oversleeping. Even my dreams were more vivid and closer to the surface of my mind so that I could remember them in exact detail.

"Knowing that this food was totally nutritious and could feed the whole planet and not create economic division or overtax the land gave it a sense of purity so I never experienced the subtle guilt feelings or reservations that might lurk in the back of the mind while eating less wholesome or ecologically sensitive foods (one acre of land can grow thousands of tons of algae or one cow per year).

"I ate the algae mixed with either fruit juice or vegetable broth (cooking water). The latter made a delicious and filling soup. I lost 15 pounds while on the fast yet never felt deprived or hungry.

"Two important aspects of the fast were drinking plenty of liquids and doing regular enemas. The enemas flushed out the old toxins and kept everything moving through. A tablespoon of bran a day mixed into the algae helped this elimination process. Drinking plenty of liquids is essential as the algae is very dry and absorbs moisture.

"I wondered, if there were a financial crash and food was hard to come by, how I would feel about living on algae forever. From the perspective of being on a regular diet like I am now I thought I might miss the variety, but I know from my experience this summer that I could do it and be satisfied and healthy."

About the Author

Christopher Hills, Ph.D., D. Sc., is founder and director of University of the Trees, Boulder Creek, California. A world renowned yogi, philosopher, and scientist, he spent many years traveling and lecturing before retiring to the giant redwoods to write. Since 1964, Dr. Hills has been President of Microalgae International Union, an organization formed to gather together those interested in feeding the world from Algae and willing to share their knowledge. In 1970 he was elected President of the World Conference on Scientific Yoga in New Delhi, India. In response to his writings, students gathered to study with him at the University of the Trees, which has evolved as a Consciousness Research School. Dr. Hills is author of many books, including *Food From Sunlight*, *Exploring Inner Space*, *Supersensomics*, *Nuclear Evolution: Discovery of the Rainbow Body*, *Rays from the Capstone*, a correspondence study course *Into Meditation Now*, *Rise of the Phoenix*, and *The Golden Egg*, a book on economic survival through the spiritualization and Common Ownership of business.

FOR FURTHER INFORMATION ON SPIRULINA:

I am interested in being placed on your mailing list for information concerning Spirulina. (Please check area of interest.)

- ☐ I wish to use it for myself.
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